From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE JEE(Advanced)-2013

ANSWERS, HINTS & SOLUTIONS
FULL TEST –I
(Paper-1)

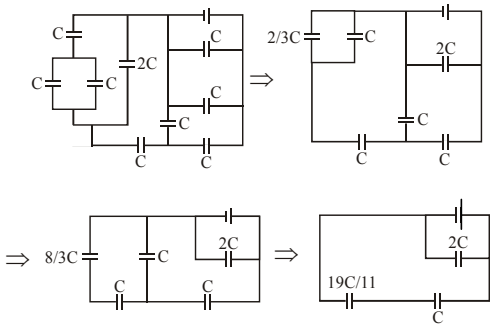
Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	D
2.	C	A	A
3.	A	B	B
4.	C	C	A
5.	D	B	D
6.	C	C	B
7.	A	D	B
8.	A	C	B
9.	C, D	A, C, D	A, C
10.	A, C	A, C	A, B
11.	A, C	A, B, C	B, C, D
12.	A, B, C, D	A, B	A, C
13.	A	B	B
14.	A	A	C
15.	C	B	D
16.	B	D	B
17.	D	A	B
18.	A	D	A
1.	(A) → (s) (B) → (p) (C) → (p) (D) → (q, r)	(A) → (p, r, t) (B) → (q) (C) → (r, s) (D) → (r, s)	(A) → (p, q, r, s) (B) → (p) (C) → (q, s) (D)→ (p, r)
2.	(A) → (r) (B) → (q) (C) → (q) (D) → (s)	(A) → (p, q, s) (B) → (p, s) (C) → (r, s) (D) → (q, t)	(A) → (s) (B) → (t) (C) → (q) (D)→ (s)

Physics PART – I

SECTION – A

1. $E^2 = V_R^2 + (V_L - V_C)^2 = V_R^2 + 0$
As, $E = V_R$
$\therefore$ $V_R = V = 220$ V
$I = \frac{E}{R} = \frac{220}{100} = 2.2$ amp

3. Equivalent circuit can be drawn as
Equivalent capacitance = $\frac{79}{30}C$.



4. This is simply discharging of a capacitor.
$\therefore$ $q = Q_0 e^{-t/CR}$ where $C = 4\pi\varepsilon_0 r$

5. When the circuit will reach to steady state the inductor will given zero resistance. Hence the entire current will only pass through it.

6. $X = \frac{H + 2H}{2} = \frac{3H}{2}$

8. Here $V \propto 1/r$
$r \rightarrow$ distance between Sun and Planet.

9. $(y - h) + \sqrt{x^2 + h^2} = \ell$

$\frac{dy}{dt} + \frac{x}{\sqrt{x^2 + h^2}}\frac{dx}{dt} = 0$

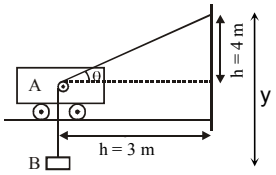
$\frac{dy}{dt} = -\frac{x}{\sqrt{x^2 + h^2}}\frac{dx}{dt}$

$\frac{dy}{dt} = -\frac{3}{5}v_A$

$|u_B| = \frac{3}{5}v_A$ …(i)

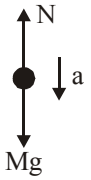
$\frac{d^2y}{dt^2} = v_A^2 \frac{h^2}{(x^2 + h^2)^{3/2}}$

$a_B = v_A^2 \frac{16}{(5)^3}$



$a_B = \frac{16}{125} v_A^2$ …(ii)

10. $Mg - T = Ma$ …(i)
$T = ma$ …(ii)
Solving (i) and (ii)
$a = \frac{Mg}{(M+m)}$
FBD of man
$Mg - N = Ma$
$N = \frac{Mmg}{(M+m)}$



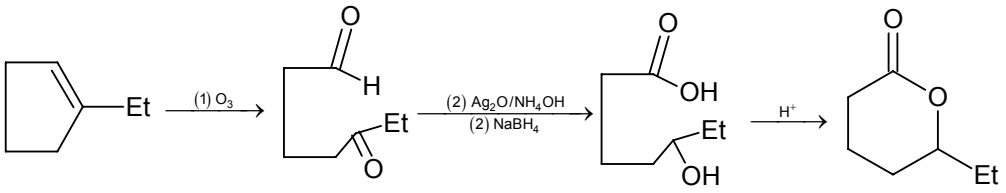
11. $T = \frac{2\pi m}{qB} = 3.14\,s.$

Chemistry

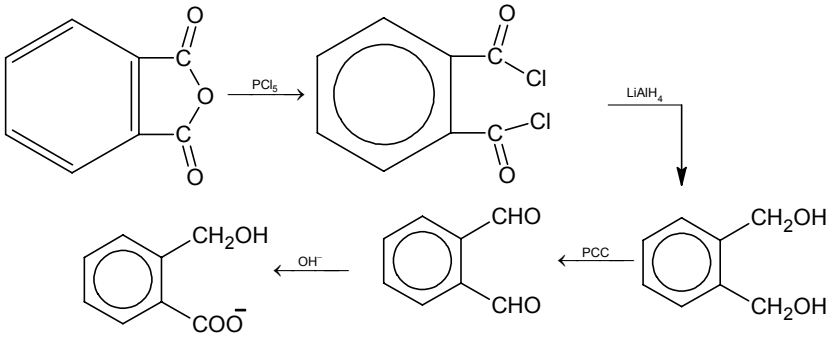
PART – II

SECTION – A

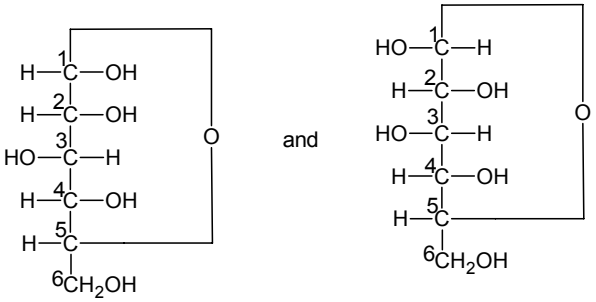
1.



2.



3.



Cyclic forms of diastereomers that differ only in the configuration of C1, are known as anomers.

The configuration differ only at carbon no. 1.

4. For equimolar solutions, $n_1 = n_2$

$\therefore x_1 = 0.5$ and 0.5

$P_B = x_B \times P_B^o = 0.5 \times 160 = 80$ mm

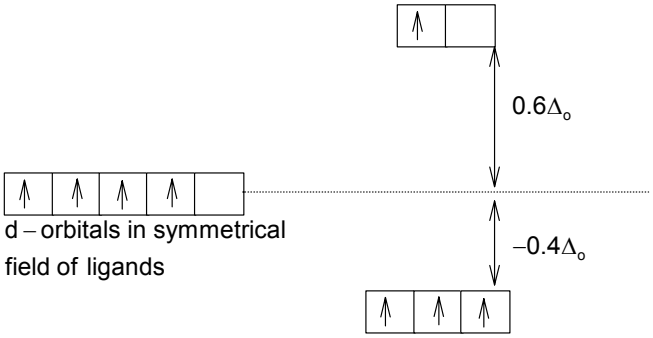
$P_T = x_T \times P_T^o = 0.5 \times 60 = 30$ mm

$P_{Total} = 80 + 30 = 110$ mm

Mole fraction of toluene in vapour phase = $\frac{P_T}{P_{Total}} = \frac{30}{110} = 0.27$

5. Entropy is a state function, i.e. the change entropy depends upon the initial and final states of the system and not on how that change is brought about.

7.



high spin d^4 configuration
in an octahedral field

CFSE = $3(-0.4)\ \Delta_o + 0.6\ \Delta_o = -1.2\Delta_o + 0.6\Delta_o = -0.6\Delta_o$

8. Given $\left(K_{sp}\right)_{AgBr} = 5\times10^{-13}$

$KBr + AgNO_3 \longrightarrow AgBr + KNO_3$

Given, $[AgNO_3] = 0.05$ M

$\Rightarrow \left[Ag^+\right] = \left[NO_3^-\right] = 0.05M$

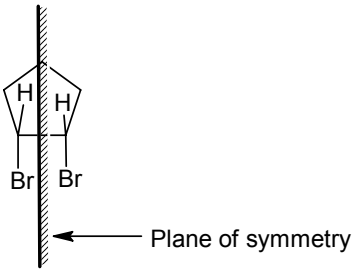
$\left[Ag^+\right]\left[Br^-\right] = \left(K_{sp}\right)AgBr \ \Rightarrow 0.05\times\left[Br^-\right] = 5\times10^{-13}$

$\Rightarrow \left[Br^-\right] = \dfrac{5\times10^{-13}}{0.05} = 1\times10^{-11}M$

$\therefore \left[Br^-\right] = \left[KBr\right] = 1\times10^{-11}M$

wt. of $KBr = 1\times10^{-11}\times120 = 1.20\times10^{-9}$ gm / litre

10.



11. (A) $Ph-CH_2-OH \xrightarrow[conc.HCl]{anh.ZnCl_2} Ph-CH_2-Cl + Zn(OH)Cl$ (immediate)

(white turbidity)

$CH_3-CH_2-OH \xrightarrow{\text{Lucas reagent}} CH_3-CH_2-Cl + Zn(OH)Cl$ (slow)

(B) $Ph-CH_2-OH \xrightarrow[\text{reagent}]{\text{Lucas}} PhCH_2Cl + Zn(OH)Cl$ (immediate)

(White turbidity)

$Ph-OH \xrightarrow[\text{reagent}]{\text{Lucas}}$ No reaction under normal condition

(C) $(CH_3)_2CH-OH \xrightarrow[\text{reagent}]{\text{Lucas}} (CH_3)_2CH-Cl + Zn(OH)Cl$ (Moderate rate)

(White turbidity)

$CH_3-CH_2-CH_2-CH_2-OH \xrightarrow[\text{reagent}]{\text{Lucas}} CH_3-CH_2-CH_2-CH_2-Cl + Zn(OH)Cl$ (Slow)

(White turbidity)

12. $H_2S + O_3 \longrightarrow H_2O + S + O_2$

$PbS + 4O_3 \longrightarrow PbSO_4 + 4O_2$

$3SO_2 + O_3 \longrightarrow 3SO_3$

$3SnCl_2 + 6HCl + O_3 \longrightarrow 3SnCl_4 + 3H_2O$

13. $Zn + Fe^{+2} \longrightarrow Fe + Zn^{+2}$

$E = E^o - \frac{0.0591}{n}\log K_{eq}$

Given, E = 0.2905

$\Rightarrow 0.2905 = E^o - \frac{0.0591}{2}\log\frac{0.01}{0.001}$

$\Rightarrow$ 0.02905 + 0.0295 log10

= 0.2905 + 0.0295

= 0.32

$\therefore E^o = \frac{0.0591}{2}\log K_{eq}$

$\Rightarrow 0.32 = \frac{0.0591}{2}\log K_{eq}$

$\Rightarrow K_{eq} = 10^{\frac{0.32}{0.0295}}$

14. For 1 mole of O_2

$O_2 \longrightarrow 2O^{2-}\left(\text{i.e. } \frac{2}{3}\times 3O^{2-}\right)$

Or $\frac{4}{3}$ mol of Al to change into Al^{+3} ions.

n = 4

Thus, $\Delta G = -nEF$

$\Rightarrow E = -\frac{\Delta G}{nF} = \frac{+827000}{4\times 96500} = 2.14$ volt

15. $E^o_{cell} = \frac{0.059}{2}\log K_{eq} \Rightarrow \frac{1.1\times 2}{0.059} = 37.2881 = \log K_{eq}$

$\Rightarrow K_{eq} = 1.9\times 10^{37}$

16. Copper sulphate when reacts with hypophosphorus acid, gets reduced to cuprous hydride.

$4CuSO_4 + 3H_3PO_2 + 6H_2O \longrightarrow 2Cu_2H_2 + 3H_3PO_4 + 4H_2SO_4$

17. $3Ba(OH)_2 + 2P_4 + 6H_2O \longrightarrow 3Ba(H_2PO_2)_2 + 2PH_3$

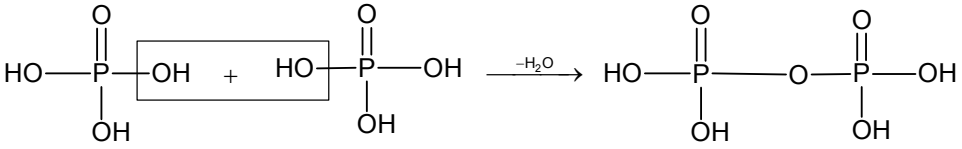
(J) (L) (K)

$Ba(H_2PO_2)_2 + H_2SO_4 \longrightarrow BaSO_4 + 2H_3PO_2$

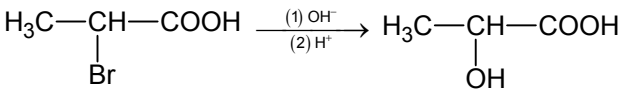
(L) (N) (M)

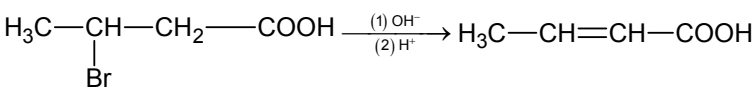
gives apple green colour in the flame

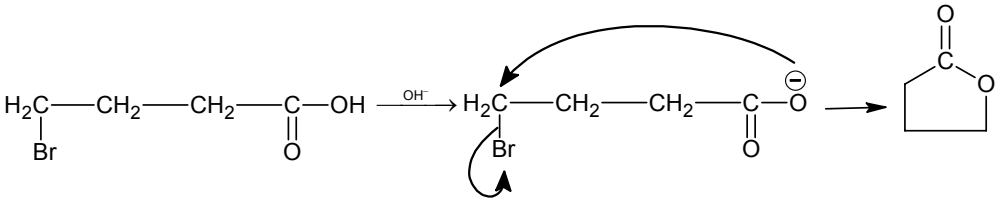
18. $H_4P_2O_7$ (pyrophosphoric acid) is a tetrabasic acid, i.e. 4-hydroxyl groups are present. It is prepared by removing one water molecule from two molecules of orthophosphoric acid. Each phosphorus atom lies in same tetrahedral environment, so called isopolyacid.

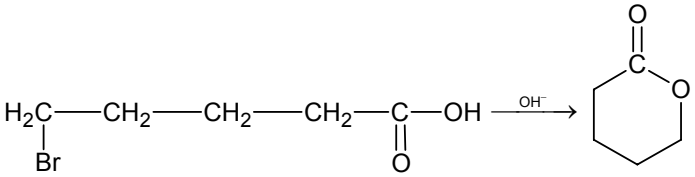


SECTION-B

1. (A) 

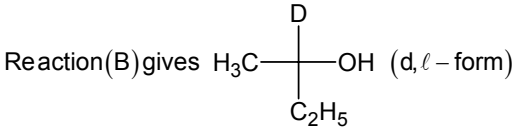
(B) 

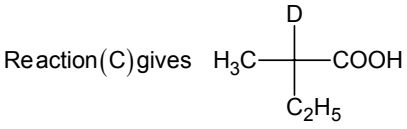
(C) 

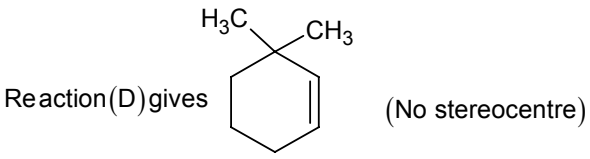
(D) 

Reactions A, C, and D show S_N2 reactions.

2. In case of (A), compound given in (A) exchanges – H with – D and becomes racemic.
In case (B), compound in (B) reacts through a carbocation and gives racemic mixture.
In case of (C), no bond attached to the stereocentre breaks in reaction of (C), so retention.
In case of (D), compound in (D) gives Hoffmann elimination and it is an example of anti-elimination. Product does not have any stereogenic centre.







Mathematics

PART – III

SECTION – A

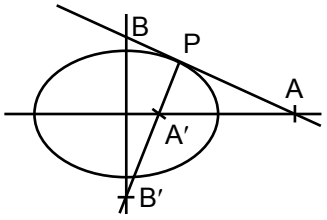
1. We must have $f(a^2 + 2a + 5) > f(a + 11)$
$\Rightarrow a^2 + 2a + 5 < a + 11$
$\Rightarrow a \in (-3, 2)$
$\therefore$ a can take values –2, –1, 0, 1

2. ΔABC is equilateral
$\Rightarrow a = b = c$

3. $AC^2 = AB^2 + BC^2$ ..... (1)
$BD^2 = BC^2 + CD^2$ ..... (2)
$CA^2 = CD^2 + DA^2$ ..... (3)
$DB^2 = DA^2 + AB^2$ ..... (4)
Equation (1) + (3) and (2) + (4) gives AC = BD
So, equation (1) and (2) give AB = CD and equation (2) and (3) give BC = AD
So Δ's CBD and ADB are congruent $\Rightarrow \angle CBD = \angle ADB = 90^\circ - \angle ABD$
$\Rightarrow \angle CBD + \angle ABD = \angle CBA \Rightarrow$ DB lies in the plane of ΔABC, so the points are coplanar

4. Let $a - d = 3t$; then $bc = (d + zt)(d + t) = d^2 + 3td + 2t^2$
Also $\frac{a}{d} = 1 + \frac{3t}{d}$, so $hk = d^2\left[1 + \frac{3t}{d}\right] = d^2t + 3td < bc$
$\Rightarrow bc > hk$

5. Clearly $P \equiv (1, 2)$
Equation of tangent at P is $4x(1) + y(2) = 8$
$\Rightarrow A \equiv (2, 0)$ and $B \equiv (0, 4)$
Similarly normal at P is $2x - 4y = 6$
$\Rightarrow A'(-3, 0)$ and $B'\left(0, \frac{3}{2}\right)$



$$\therefore \text{Area } BPB' = \frac{1}{2}\begin{vmatrix} 0 & 4 & 1 \\ 1 & 2 & 1 \\ 0 & \frac{3}{2} & 1 \end{vmatrix} = \frac{5}{4} \text{ and Area } APA' = 5$$

$\therefore$ Ratio = 4

6. As $\sin x + \sin y \geq \cos\alpha \cos x \ \forall\ x \in R$
Let $x = -\frac{\pi}{2} \Rightarrow \sin y \geq 1 \Rightarrow \sin y = 1$
$\Rightarrow 1 + \sin x \geq \cos\alpha \cos x$
$\Rightarrow \cos\alpha \cos x - \sin x \leq 1$
$\Rightarrow \sqrt{\cos^2\alpha + 1} = 1 \Rightarrow \cos\alpha = 0 \Rightarrow \sin y + \cos\alpha = 1 + 0 = 1$

7. $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx < \int_0^1 \frac{x}{\sqrt{x}} dx = \frac{2}{3}$

$\Rightarrow I < \frac{2}{3}$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} < \int_0^1 x^{-1/2} dx = 2$

$\Rightarrow J < 2$

8. Equation of tangent $x + t^2 y = 2ct \Rightarrow T \equiv (2ct, 0)$ and $T' \equiv \left(0, \frac{2c}{t}\right)$

Equation of normal $t^3 x - ty = c(t^4 - 1)$; $N \equiv \left[c\left(t - \frac{1}{t^3}\right), 0\right]$; $N \equiv \left[0, c\left(\frac{1}{t} - t^3\right)\right]$

$\Rightarrow NT = \frac{c}{t^3} + ct$; $N'T' = ct^3 + \frac{c}{t}$

$\Rightarrow \Delta = \frac{1}{2}\left|\frac{c}{t^3} + ct\right|\left|\frac{c}{t}\right| \Rightarrow \frac{1}{\Delta} = \frac{2t^4}{c^2(1+t^4)}$ and $\Delta' = \frac{1}{2}\left|ct^3 + \frac{c}{t}\right|ct$

$\Rightarrow \frac{1}{\Delta'} = \frac{2}{c^2(1+t^4)} \Rightarrow \frac{1}{\Delta} + \frac{1}{\Delta'} = \frac{2}{c^2}$

9. We have $2\sin^2\left(\frac{\pi}{2}\cos^2 x\right) = 2\sin^2\left(\frac{\pi}{2}\sin 2x\right)$

$\Rightarrow \cos^2 x = \sin 2x \Rightarrow \cos x(\cos x - 2 \sin x) = 0$

$\Rightarrow 1 - 2 \tan x = 0$ as $\cos x \neq 0$

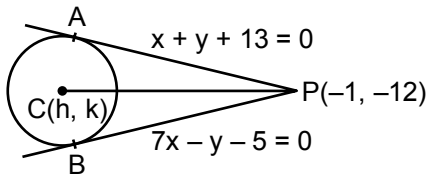
$\Rightarrow \tan x = \frac{1}{2}$ and $\cos 2x = \frac{3}{5}$

10. Clearly, the point lies on $7x - y = 5$

Also, centre of the circle must lie on the bisectors of the lines

$x + y + 13 = 0$ and $7x - y - 5 = 0$ given by

$\frac{x + y + 13}{\sqrt{2}} = \pm\frac{7x - y - 5}{\sqrt{50}} \Rightarrow x - 3y = 35$ and $3x + y + 15 = 0$



Let (h, k) be the centre of the circle, then $h - 3k = 35$ ….. (1)

and $3h + k = 15$ ….. (2)

Clearly CB is perpendicular BP $\Rightarrow \frac{k-2}{h-1} \times 7 = -1 \Rightarrow h + 7k - 15 = 0$ ….. (3)

On solving, we get centres as $C_1 \equiv (29 - 2)$ and $C_2 \equiv (-6, 3)$

$\Rightarrow r_1^2 = 800$ and $r_1^2 = 50 \Rightarrow$ smaller circle has radius = $\sqrt{50}$

Therefore area of quadrilateral ACBP = $2\left[\frac{1}{2} \times \sqrt{50} \times \sqrt{200}\right]$ sq. units

11. Let $f(x) = ax^2 + bx + c$ and $f(1) = a + b + c < 0 \Rightarrow f(x) < 0$ for all $x \in R \Rightarrow f(-2) < 0 \Rightarrow 4a - 2b + c < 0$

Again $f(3) + f(2) < 0 \Rightarrow 13a + 5b + 2c < 0$

Also $f(-5) < 0 \Rightarrow 25a - 5b + c < 0 \Rightarrow 5b - 25a - c > 0$

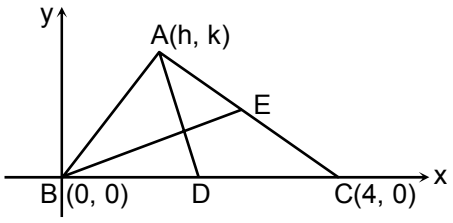
12. $f(x) = \begin{cases} (x-x)(-x) = 0 & ; \quad x < 0 \\ (x+x)x = 2x^2 & ; \quad x \geq 0 \end{cases}$

$\Rightarrow$ f is continuous and differentiable

Again $f'(x) = \begin{cases} 0 & ; \quad x < 0 \\ 4x & ; \quad x \geq 0 \end{cases}$

$\Rightarrow$ f′(x) is continuous but not differentiable at x = 0

13. Taking B as origin, BC as x–axis and A as (h, k) and C(4, 0)



We have D ≡ (2, 0) and E ≡ $\left(\frac{h+4}{2}, \frac{k}{2}\right)$

$\Rightarrow K^2 + (h + 4)(h - 2) = 0$ ..... (1)

Also AC = 3 $\Rightarrow (h-4)^2 + k^2 = 9$ ..... (2)

And area = $\frac{1}{2} \times 4 \times K = 2K$

Solving (1) and (2), we get $h = \frac{3}{2}$ and $K = \frac{\sqrt{11}}{2}$

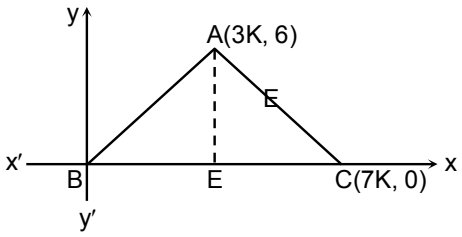
$\therefore \Delta ABC = \sqrt{11}$

14. Let AD be angle bisector $\Rightarrow \frac{AB}{AC} = \frac{BD}{DC} \Rightarrow c = 2b$

Now b + c > a $\Rightarrow$ b + c > 6 $\Rightarrow$ b + 2b > 6 $\Rightarrow$ b > 2

Again $\frac{b^2 + 4b^2 - 6}{4b^2} < 1 \Rightarrow b < 6 \Rightarrow b \in (2, 6)$ and consequently $c \in (4, 12)$

15. $\frac{BE}{EC} = \frac{3}{4}$, take BE = 3K and EC = 4K



$\Rightarrow$ A must be (3K, 6)

Now equation of line BA is $y = \frac{6}{3K}x$

$\Rightarrow Ky - 2x = 0$

Since perpendicular from c on AB = 7

$\Rightarrow \left|\frac{0 - 14K}{\sqrt{K^2 + 4}}\right| = 7 \Rightarrow K^2 = \frac{4}{3}$

Now $c^2 = (6-0)^2 + (3K-0)^2$ = 36 + 9K² = 36 + 12

$\Rightarrow c = \sqrt{48} = 4\sqrt{3}$

16–18. Let $12\int_0^1 yf(y)dy = a$ and $20\int_0^1 \left(y^2 f(y)\right)dy + 4 = b$

$\Rightarrow f(x) = ax^2 + bx \Rightarrow f(y) = ay^2 + by$

$\Rightarrow f(x) = 12x^2\int_0^1 y\left(ay^2 + by\right)dy + 20x\int_0^1 y^2\left(ay^2 + by\right)dy + 4x = 12x^2\left(\frac{a}{4} + \frac{b}{3}\right) + 20x\left[\frac{a}{5} + \frac{b}{4}\right] + 4x$

$\Rightarrow$ a = 3a + 4b $\Rightarrow$ a = –2b ..... (1)

and 4a + 5b + 4 = b $\Rightarrow$ a + b = –1 ..... (2)

$\Rightarrow$ a = –2, b = 1

$\therefore f(x) = -2x^2 + x$

SECTION – B

1. Clearly we have $f(x)=\begin{cases}-2 & ; \quad x\in[-2,\ 1)\\ -1 & ; \quad x\in[-1,\ 0)\\ x & ; \quad x\in[0,\ 2]\end{cases}$ and g(x) = sec x; $x \in R - (2x+1)\frac{\pi}{2}$

We have $fog(x)=\begin{cases}-2 & ; \quad \sec x\in[-2,\ -1)\\ -1 & ; \quad \sec x\in[-1,\ 0)\\ \sec x & ; \quad \sec x\in[0,\ 2]\end{cases}$

$\therefore\ fog(x)=\begin{cases}-2 & ; \quad x\in\left[-\frac{4\pi}{3},\ -\frac{2\pi}{3}\right]\cup\left[\frac{2\pi}{3},\ \frac{4\pi}{3}\right]-\{-\pi,\ \pi\}\\ -1 & ; \quad x=-\pi,\ \pi\\ \sec x & ; \quad x\in\left[-\frac{\pi}{3},\ \frac{\pi}{3}\right]\end{cases}$

Limit of fog exist at $x = -\pi, \pi, -1$, points of discontinuity of fog are $-\pi, \pi$ and points of differentiability of fog are $-1, \frac{5\pi}{6}$

Again $g(f(x))=\begin{cases}\sec(-2) & ; \quad x\in[-2,\ -1)-\left\{-\frac{\pi}{2}\right\}\\ \sec(-1) & ; \quad x\in[-1,\ 0)\\ \sec x & ; \quad x\in[0,\ 2]-\left\{\frac{\pi}{2}\right\}\end{cases}$

Limit of g(f(x)) does not exist at x = –1

2. (A) Let E_i ; i = 1, 2, 3, 4 represents the events that the bag contains I white balls: Clearly $P(E_i)=\frac{1}{4}$. Let W be the event that the ball drawn is white then,

$$P(W)=\sum P(E_i)\cdot P\left(\frac{W}{E_i}\right)=\frac{1}{4}\left(\frac{1}{4}+\frac{2}{4}+\frac{3}{4}+\frac{4}{4}\right)=\frac{5}{8}$$

Now $P\left(\frac{E_4}{W}\right)=\frac{P(E_4)P(W/E_4)}{P(W)}=\frac{1/4}{5/8}=\frac{2}{5}=\frac{p}{15}\ \Rightarrow p = 6$

(B) Required number of ways

$$= {}^{12}C_1+{}^{12}C_2\left({}^2C_1+2\cdot{}^2C_2\right)+{}^{12}C_3\left({}^3C_1+2\cdot{}^3C_2\right)+.....+{}^{12}C_{12}\left({}^{12}C_1+2\cdot{}^{12}C_2\right)$$

$$=\left({}^{12}C_1+2\cdot{}^{12}C_2+3\cdot{}^{12}C_3+.....+12\cdot{}^{12}C_2\right)+2\left({}^{12}C_2\cdot{}^2C_2+{}^{12}C_3\cdot{}^3C_2+.....+{}^{12}C_2\cdot{}^{12}C_2\right)$$

$$=\sum_{r=1}^{12} r\cdot{}^{12}C_r+12\sum_{r=2}^{12}{}^{10}C_{r-2}=12\times2^{11}+12\times11\times2^{10}$$

$= 13 \times 2^{10} \times 12 = 13 \times 2^{10} \times m \Rightarrow m = 12$

(C) $$\frac{5x}{2-x}+\frac{5y}{2-y}+\frac{5z}{2-z}=5\left[\frac{x}{2-x}+\frac{y}{2-y}+\frac{z}{2-z}\right]$$

$$=5\left[\frac{x-2+2}{2-x}+\frac{y-2+2}{2-y}+\frac{z-2+2}{2-z}\right]=5\left[-3+2\left[\frac{1}{2-x}+\frac{1}{2-y}+\frac{1}{2-z}\right]\right]$$

Now apply A.M ≥ H.M on 2 – x, 2– y and 2 – z

$$\Rightarrow \frac{2-x+2-y+2-z}{3} \geq \frac{3}{\frac{1}{2-x}+\frac{1}{2-y}+\frac{1}{2-z}} \Rightarrow \frac{1}{2-x}+\frac{1}{2-y}+\frac{1}{2-z} \geq \frac{9}{5}$$

Hence $\frac{5x}{2-x}+\frac{5y}{2-y}+\frac{5z}{2-z} \geq 5\left[-3+2\cdot\frac{9}{5}\right] = 3$

$\therefore$ Least value = 3

(D) $\sum_{K=1}^{n} 12\cdot K\ {}^{12}C_K \cdot\ {}^{11}C_{K-1} = 12^2 \sum_{K=1}^{12}\left({}^{11}C_{K-1}\right)^2 = 12^2 \cdot \frac{22!}{11!\ 11!}$

$= 12\cdot\frac{21\cdot 19\cdot 17.....3}{11!} - 2^{12}\cdot 6 \Rightarrow p = 6$